

Hieu Tran

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RESEARCH INTERESTS

My primary research interest is enabling LLMs to understand, generate, and reason about human language to enhance their ability to perform complex cognitive and language-related tasks. Recently, I have expanded my work to advanced topics such as Reinforcement Learning and Test Time Computing, with a particular emphasis on improving the reasoning capabilities, factual accuracy, and efficiency of language models in both general and domain-specific applications.

EDUCATION

University of Massachusetts, Amherst (UMass)

Computer Science PhD student

- GPA: 3.94/4.0

Aug. 2022 - now

Massachusetts, USA

Hanoi University of Science and Technology (HUST)

Engineer's Degree in Computer Science

- GPA: 3.6/4.0 (top 1.3%), graduated with Excellent Degree

Aug. 2014 - Aug. 2019

Hanoi, Vietnam

Bien Hoa High School for The Gifted

Specialized in Informatics

Sep. 2011 - Sep. 2014

Hanam, Vietnam

RESEARCH EXPERIENCE

Computer Science PhD student

BioNLP lab (<https://bio-nlp.org>)

- Advisor: Prof. Hong Yu ([google scholar](#))
- Research Topic: Large Language Models for Medical

Aug 2022 - now

Massachusetts, USA

Research Intern

United Imaging Intelligence (<http://www.uui-ai.com>)

- Supervisor: Dr. Weijing Huang ([google scholar](#))
- Working on fact-checking of LLMs responses

May 2024 - Aug 2024

Boston, USA

AI Research Resident

VinAI Research (www.vinai.io)

- Supervisor: Prof. Thien Huu Nguyen ([google scholar](#))
- Research Topic: Information Extraction

Nov 2019 - Jun 2022

Hanoi, Vietnam

Applied Rotation Program

VinAI Research

- Supervisor: Dr. Duy Tin Vo ([google scholar](#)), Dr. Dat Quoc Nguyen ([google scholar](#))
- Working on knowledge-based question answering and machine translation problem

May 2021 - Sep 2021

Hanoi, Vietnam

PUBLICATIONS

EXPLOITING TREE STRUCTURE FOR CREDIT ASSIGNMENT IN REINFORCEMENT LEARNING WITH LARGE LANGUAGE MODELS

2025

Hieu Tran, Zonghai Yao, Hong Yu

Proceedings of ACL 2026 (Findings)

PRIME: Planning and Retrieval-Integrated Memory for Enhanced Reasoning

2025

Hieu Tran, Zonghai Yao, Nguyen Luong Tran, Zhichao Yang, Feiyun Ouyang, Shuo Han, Razieh Rahimi, Hong Yu

Proceedings of AAAI 2026

LEAF: Learning and Evaluation Augmented by Fact-Checking to Improve Factualness in Large Language Models

2025

Hieu Tran, Junda Wang, Yujan Ting, Hong Yu, Weijing Huang, Terrence Chen

Proceedings of EMNLP 2025 (Industry Track)

Chatbot To Help Patients Understand Their Health

2025

- Won Seok Jang*, **Hieu Tran***, Manav Mistry, SaiKiran Gandluri, Yifan Zhang, Sharmin Sultana, Sunjae Kwon, Zonghai Yao, Hong Yu
 Proceedings of EMNLP 2025 (Findings) 2025
- RARE: Retrieval-Augmented Reasoning Enhancement for Large Language Models**
Hieu Tran, Zonghai Yao, Zhichao Yang, Junda Wang, Yifan Zhang, Shuo Han, Feiyun Ouyang, Hong Yu
 Proceedings of ACL 2025
- BioInstruct: Instruction Tuning of Large Language Models for Biomedical Natural Language Processing**
Hieu Tran, Zhichao Yang, Zonghai Yao, Hong Yu
 Journal of the American Medical Informatics Association, ocae122 2024
- Readme: Bridging medical jargon and lay understanding for patient education through data-centric nlp**
 Zonghai Yao, Nandyala Siddharth Kantu, Guanghao Wei, **Hieu Tran**, Zhangqi Duan, Sunjae Kwon, Zhichao Yang, Hong Yu
 Proceedings of EMNLP 2024 (Findings) 2023
- A Vietnamese-English Neural Machine Translation System**
 Thien Hai Nguyen, Tuan-Duy H. Nguyen, Duy Phung, Duy Tran-Cong Nguyen, **Hieu Minh Tran**, Manh Luong, Tin Duy Vo, Hung Hai Bui, Dinh Phung, Dat Quoc Nguyen
 Proceedings of InterSpeech 2022 - Show & Tell Demonstrations 2022
- Exploiting Document Structures and Cluster Consistencies for Event Coreference Resolution**
Hieu Tran, Duy Phung, Thien Huu Nguyen
 Proceedings of ACL-IJCNLP 2021 (oral presentation) 2021
- Learning Cross-lingual Representations for Event Coreference Resolution with Multi-view Alignment and Optimal Transport**
 Duy Phung, **Hieu Tran**, Thien Huu Nguyen
 Proceedings of the first Workshop on Multilingual Representation Learning (MRL 2021) at EMNLP 2021 2021
- Vietnamese Speech-based Question Answering over Car Manuals**
 Tin Duy Vo, Manh Tien Luong, Duong Minh Le, **Hieu Tran**, Nhan Tri Do, Duy Nguyen, Thien Hai Nguyen, Hung Hai Bui, Dat Quoc Nguyen, Dinh Quoc Phung
 Proceedings of Data-Centric AI workshop NeurIPS 2021 2021
- The Dots Have Their Values: Exploiting the Node-Edge Connection in Graph-based Neural Models for Document-level Relation Extraction**
Hieu Tran, Minh Trung Nguyen, Thien Huu Nguyen
 Proceedings of EMNLP 2020 (Findings) 2020

HONORS AND AWARDS

- Optum AI Research Fellowship** 2025
 Optum AI
- The Excellence Scholarship - Level A** 2015
 School of Information and Communication Technology, HUST
 Each semester top 5% of students with greatest academic performance are awarded
- National Excellent Student Award** 2014
 Vietnam Ministry of Education and Training
 Third prize in Informatics subject, National Excellent High School Student Contest

ACTIVITIES

- Technical Talk** August 2021
 AI Day 2021: Empowering Innovations – website: <https://www.vinai.io/aiday2021>
 Present at AI Day 2021 - the event which welcomed technical talks from top researchers around the world and attracted over 20,000 views online

CERTIFICATES

- Deep Learning Specialization by deeplearning.ai**
- Neural Networks and Deep Learning
 - Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization
 - Structuring Machine Learning Projects
 - Convolutional Neural Networks
 - Sequence Models